👩‍🏫 Staff Guide · Biology A · Orcas & the Salish Sea (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A1 J32 + A2 Killer whale webquest	Reading parts 1 and 5; notes section A
A3 Salish Sea food web	Notes food-web table
A4 Trace the Toxin + A5 PCB articles	Reading part 3; notes section B
A6 Ocean noise + A7 Necropsy	Reading part 4; notes section C
A9 Orca review + Orca Test	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

🐋 Orcas & the Salish Sea: Guided Notes

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Use the class slides and your reading. Key words:

Southern Residents · *pod* · *matriline* · *food web* · *producer* · *PCBs* · *biomagnification* · *blubber* · *echolocation*

A. The Southern Residents (A1, A2)

1. The three Southern Resident pods are J, K and _____.
2. A mother and her young are a _____.
3. Transient (Bigg's) orcas eat seals; Residents eat Chinook _____.

B. Food web and toxins (A3, A4, A5)

4. Fill in the table.

Level	Example	What it eats
Producer		(makes food from sunlight)
Primary consumer	zooplankton	
Top consumer		

5. Arrows in a food web show the flow of _____.
6. PCBs are stored in _____ because they are fat-soluble.
7. Why do orcas have more PCBs than salmon?

C. Noise and solutions (A6, A7)

8. Orcas find fish with _____.
9. How can boat noise lead to more toxins in an orca's blood?
10. Pick one solution and explain how it helps orcas.

🐋 Orcas & the Salish Sea: Reading

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Why Are the Southern Residents in Trouble?

The Southern Resident killer whales (SRKW) live in the Salish Sea, the waters between Washington and British Columbia. They travel in three pods, J, K and L, and there are fewer than 80 of them. Orca families are close: a mother and her young form a matriline, related families form a pod, and pods that share calls form a clan.

Southern Residents eat mostly Chinook salmon. That puts them at the top of a long food web. Phytoplankton are producers: they make food from sunlight. Zooplankton eat phytoplankton, herring eat zooplankton, salmon eat herring, and orcas eat salmon. The arrows in a food web show energy moving from one living thing to the next.

PCBs are toxic chemicals that wash into the sea from old industry and storm drains. PCBs are fat-soluble, so they are stored in fat, like orca blubber. Each animal eats many animals from the level below, so the toxin becomes more concentrated at every step. This is called biomagnification, and it means orcas carry the most.

Orcas hunt with echolocation: they make clicks and listen for the echoes that bounce off fish. Boat and ship noise covers those echoes, so hunting takes more time and energy. When orcas cannot find enough food, they burn their blubber, and the stored PCBs are released into their blood.

In 2018 the orca J32 died while carrying a calf. Scientists found that lack of food, toxins and noise worked together. Solutions include restoring salmon rivers (like removing the Elwha dams), slowing ships, keeping toxins out of storm drains, and planting trees to shade salmon streams.

Check your reading

1. What do Southern Resident orcas mostly eat?

2. Explain biomagnification in your own words.

3. Name the three threats that worked together in J32's death.

🐋 Orcas & the Salish Sea: Quick Check

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Phytoplankton are…

A. Producers | B. Top consumers | C. Toxins

2. Arrows in a food web show…

A. Where animals sleep | B. Ocean currents | C. The flow of energy

3. PCBs are stored in orca…

A. Teeth | B. Fins | C. Blubber (fat)

4. Echolocation means orcas…

A. Swim in circles | B. Use clicks and echoes to find fish | C. Sleep near the surface

5. Put this food chain in order: salmon, phytoplankton, orca, zooplankton, herring.

6. Give one way people can help the Southern Residents.

🐋 Orcas & the Salish Sea: Guided Notes · ANSWER KEY

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Use the class slides and your reading. Key words:

Southern Residents · *pod* · *matriline* · *food web* · *producer* · *PCBs* · *biomagnification* · *blubber* · *echolocation*

A. The Southern Residents (A1, A2)

1. The three Southern Resident pods are J, K and **L**.
2. A mother and her young are a **matriline**.
3. Transient (Bigg's) orcas eat seals; Residents eat Chinook **salmon**.

B. Food web and toxins (A3, A4, A5)

4. Fill in the table.

Level	Example	What it eats
Producer	**phytoplankton**	(makes food from sunlight)
Primary consumer	zooplankton	**phytoplankton**
Top consumer	**orca**	**Chinook salmon**

5. Arrows in a food web show the flow of **energy**.
6. PCBs are stored in **blubber** because they are fat-soluble.
7. Why do orcas have more PCBs than salmon?
Each level eats many of the level below, so PCBs build up (biomagnify); orcas are at the top.

C. Noise and solutions (A6, A7)

8. Orcas find fish with **echolocation**.
9. How can boat noise lead to more toxins in an orca's blood?
Noise makes hunting harder, so the orca gets less food and burns blubber, which releases stored PCBs.
10. Pick one solution and explain how it helps orcas.
Example: restoring rivers / removing dams brings back salmon, so orcas have more food.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

🐋 Orcas & the Salish Sea: Reading · ANSWER KEY

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

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The Southern Resident killer whales (SRKW) live in the Salish Sea, the waters between Washington and British Columbia. They travel in three pods, J, K and L, and there are fewer than 80 of them. Orca families are close: a mother and her young form a matriline, related families form a pod, and pods that share calls form a clan.

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Orcas hunt with echolocation: they make clicks and listen for the echoes that bounce off fish. Boat and ship noise covers those echoes, so hunting takes more time and energy. When orcas cannot find enough food, they burn their blubber, and the stored PCBs are released into their blood.

In 2018 the orca J32 died while carrying a calf. Scientists found that lack of food, toxins and noise worked together. Solutions include restoring salmon rivers (like removing the Elwha dams), slowing ships, keeping toxins out of storm drains, and planting trees to shade salmon streams.

Check your reading

1. What do Southern Resident orcas mostly eat?
Chinook salmon.

2. Explain biomagnification in your own words.
Toxins get more concentrated at each step up the food chain, so top predators like orcas have the most.

3. Name the three threats that worked together in J32's death.
Lack of food (salmon), toxins (PCBs), and noise.

🐋 Orcas & the Salish Sea: Quick Check · ANSWER KEY

Biology A · Unit 1 · Level 3 (closer to the class work) · goes with Unit 1 Orca: A1 J32, A2 webquest, A3 food web, A4 Trace the Toxin, A5 PCB articles, A6 Ocean noise, A9 Orca Test

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Phytoplankton are…

A. Producers | B. Top consumers | C. Toxins

2. Arrows in a food web show…

A. Where animals sleep | B. Ocean currents | **C. The flow of energy**

3. PCBs are stored in orca…

A. Teeth | B. Fins | **C. Blubber (fat)**

4. Echolocation means orcas…

A. Swim in circles | **B. Use clicks and echoes to find fish** | C. Sleep near the surface

5. Put this food chain in order: salmon, phytoplankton, orca, zooplankton, herring.

Phytoplankton → zooplankton → herring → salmon → orca.

6. Give one way people can help the Southern Residents.

Any: restore salmon/remove dams, slow ships, keep toxins out of drains, plant trees by streams.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*